Amaka Enueme, PhD

enuemeamaka@gmail.com | LinkedIn | 424.414.9654
Los Angeles, CA, 90064

Education

University of California, Los Angeles (UCLA) Los Angeles, CA
PhD, Bioengineering June 2025

- **Dissertation:** Biomaterials for Enhanced Care in Sickle Cell Disease and Complex Wounds
- **Committee:** Dr. Andrea M. Kasko (Chair), Dr. Nasim Annabi, Dr. Nicholas M. Bernthal, Dr. Timothy J. Deming, Dr. Nabil J. Tawil

University of California, Los Angeles (UCLA) June 2022
MSc, Bioengineering

Covenant University Ota, Nigeria
BSc, Microbiology July 2019

Research Interests

Materials engineering • cell-material interactions • polymer chemistry • wound biology • controlled drug delivery • biomaterials • translational therapeutics

Teaching Experience

UCLA Los Angeles, CA
Teaching Assistant, Department of Bioengineering Mar 2025 – June 2025
Targeted drug delivery and controlled drug release, BE 183/283; Spring 2025

- Led weekly discussion sections for undergraduate and graduate students focused on critical analysis of peer-reviewed scientific literature
- Developed a teaching module based on my doctoral research demonstrating how real-time research can be translated into real-world applications
- Designed grading rubrics for written assignments and oral presentations, emphasizing clarity in scientific reasoning and communication
- Facilitated student engagement through discussion-based and active learning approaches

Teaching Assistant, Department of Life Sciences Jan 2022 – Mar 2022
Genetics, LS 107; Winter 2022

- Delivered lectures to 50+ students using visual models and problem-solving approaches to teach Mendelian genetics, inheritance, genetic analysis, and human genetic concepts
- Designed lesson plans and taught classes 3 times a week with a team of 2 graduate colleagues
- Applied active learning strategies to support students with diverse academic backgrounds

Teaching Assistant, Department of Molecular, Cell and Developmental Biology Jan 2021 – June 2021
Developmental biology, MCDB 138; Winter 2021, Spring 2021

- Led discussion sections for 100+ students focused on experimental design, cell and developmental biology, data interpretation, and statistical analysis
- Collaborated with a team of 3 graduate student colleagues to design exams and analyze scientific writing with students.
- Supported students through structured discussions that reinforced complex biological concepts

Perfecta Montessori School Asaba, Nigeria
Science Director Aug 2019 – Dec 2020

- Designed and taught engaging science lessons rooted in the Montessori principles, covering foundational topics in DNA, chemistry, and physics
- Developed hands-on activities and age-appropriate curriculum for 30+ students across two grade levels
- Led a transition to virtual instruction during the COVID-19 pandemic while maintaining student participation

Research Experience

Handshake AI
Materials Science Fellow Aug 2025 – Present

- Designed and evaluated advanced biology and materials science research problems to improve AI scientific reasoning and accuracy
- Collaborated with interdisciplinary teams of chemists, biologists, and AI researchers to develop technical research workflows

UCLA Los Angeles, CA
Graduate Researcher Sep 2020 – June 2025

Project: Biomaterials for Enhanced Care in Sickle Cell Disease and Complex Wounds

- Designed and executed mechanical characterization methods using ASTM testing standards with Instron and DMA technologies to evaluate the performance of synthesized biomaterials
- Designed, synthesized, and characterized biodegradable polymer formulations using organic chemistry principles for controlled drug delivery and wound healing, contributing to a U.S. provisional patent application
- Advanced research findings with a multidisciplinary team of 6 community college interns and 2 undergraduate researchers, fostering cross-level mentorship and team-driven discovery
- Communicated research results to diverse groups including faculty, colleagues, collaborators and undergrad mentees, through written weekly reports and monthly presentations that provided an in-depth discussion of scientific findings

UCLA and UTMB CA and TX
Translational Researcher Apr 2021 – June 2025

Project: Translational Hydrogel Drug Delivery Systems for Battlefield Wound Management

- Advanced a DoD-funded drug delivery platform from computational design and bench-top development through preclinical studies (ovine models) for traumatic wound management
- Led the materials characterization and scale-up of therapeutic biomaterials from in vitro design studies through small- and large-animal models
- Collaborated and advised on project progression through monthly meetings with cross-functional partners, including surgeons, bioengineers, medical doctors, and military stakeholders
- Validated research progress with key stakeholders from the U.S. DoD by preparing detailed quarterly reports and proposals delivered to my project team and collaborators, ensuring continued funding

National Institute for Pharmaceutical Research and Development Abuja, Nigeria
Microbiology Intern Feb 2018 – July 2018

- Performed CD4 and viral load diagnostics on 200+ HIV-positive patients using flow cytometry, working with infectious human material under BSL-2 conditions with results used to inform patient treatment plans
- Trained and supervised 5 junior interns; authored 3 regulatory protocols to standardize documentation and ensure knowledge transfer to incoming team members

Covenant University Ota, Nigeria
Undergraduate Researcher Jan 2017 – July 2019

- Characterized bacterial isolates from 30+ human samples using culture and biochemical identification, and investigated antibiotic-resistance plasmids through plasmid isolation, PCR, and gel electrophoresis

Mentorship and Student Development

UCLA Department of Bioengineering Los Angeles, CA
Research mentor Jan 2022 – June 2025

- Led and mentored a team of 8, empowering them to take ownership of key project components and accelerating the progress of 3 high-impact research projects
- Fostered inclusive learning environments through collaborative problem-solving and experiential learning
- Mentored undergraduate researchers in laboratory curriculum: experimental design, laboratory techniques, data analysis, scientific communication, and independent research, with former mentees progressing to graduate training at UC Berkeley and UC Davis

Peer-Reviewed Publications

Pumford, E. A., Hamad, C., **Enueme, A.I**., Mamouei, Z., Bernthal, N.M., Kasko, A.M., Wenke, J.C., Wainwright, W. Advanced drug eluting hydrogel wound dressing for prolonged field care: Innovations in drug delivery and trauma management. ***Scientific Reports.*** https://doi.org/10.1038/s41598-026-37521-y

Jackson Hoffman, B. A., Pumford, E. A., **Enueme, A. I**., Fetah, K. L., Friedl, O. M., & Kasko, A. M. (2023). Engineered macromolecular Toll-like receptor agents and assemblies. ***Trends in Biotechnology***, *41*(9), 1139–1154

Akinnola, O.O., Ajayi, S.A., Ogunleye, B.O., **Enueme, N.I**. (2020). Disinfection by-products in swimming pools and health-related issues. In M.N.V Prasad (Eds.), ***Disinfection Byproducts in Drinking Water*** (pp. 235-252). Elsevier Science

Intellectual Property

Enueme, A.I., Kasko, A.M. (2026). Multifunctional Chitosan Hydrogel Wound Dressing for Sustained Antimicrobial and Vasodilatory Therapy in Sickle Cell Ulcers – U.S Provisional U.S. Patent. Case No. 2026-019-1

Manuscript in Preparation

Enueme, A.I., Kasko, A.M. Multifunctional chitosan hydrogels targeting sickle cell leg ulcers. ***Journal of Biomedical Materials Research Part A***. Manuscript in preparation

Presentations

Enueme, A. I., Kasko, A. M. (2025, June). Chitosan Hydrogels for Treating Venous Ulcers. Poster presentation at Amgen University Day, **Amgen**

Enueme, A. I., Kasko, A. M. (2025, March). Chitosan Hydrogels for Treating Venous Ulcers. Poster presentation at Excellence in Graduate Polymer Research Symposium, ***American Chemical Society***, National Meeting.

Enueme, A. I., Kasko, A. M. (2024, February). Chitosan Hydrogels for Treating Venous Ulcers. Oral presentation at 7th Annual Bioengineering Research Day, **University of California, Los Angeles**

Enueme, A. I. (2018, November). Coagulase Negative Staphylococci. Seminar presentation to the Biological Science Department, **Covenant University**, Nigeria

Honors & Awards

ACS Excellence in Graduate Polymer Research Award	Mar 2025
UCLA Dissertation Year Award | $30,000	June 2024
UCLA Bioengineering Research Day Award	Feb 2024
Best Graduating Student, Microbiology	July 2019
Best Graduating Student, Department of Biological Sciences	July 2019

Leadership and Outreach Experience

American Chemical Society (Southern California Section) Los Angeles, CA
Executive Committee Member-at-Large Jan 2026 – Present

- Contribute to executive committee discussions and initiatives supporting scientific programming, professional development, and engagement among chemists across Southern California

UCLA Graduate Student Association Sep 2022 – June 2023
Director of Internal Affairs

- Led a student housing resolution initiative for 14,000+ students which was reviewed by the UC Office of President

Director of Discretionary Funds Sep 2021 – June 2022; Sep 2023 – June 2024

- Partnering with a team of graduate students colleagues to manage council expenditures across 130+ academic departments

Mercy Worldwide | CARE Mar 2022 – Present
Community Outreach Ambassador

- Partnered with the LA Food Bank to package and distribute essential groceries to underserved populations, reaching over 900,000 individuals monthly
- Participated in senior home outreach programs and annual toy drives

UCLA Women in Engineering Leadership Academy Oct 2020 – Feb 2021

- Selected as one of 36 participants for a competitive leadership academy focused on empowering women in STEM fields
- Delivered a capstone presentation on navigating and succeeding in leadership roles within academic and professional settings

Teaching & Technical Skills

- Teaching & Mentorship: lesson planning, active learning, curriculum design, classroom instruction, mentorship, inclusive pedagogy
- Research techniques: Organic synthesis, polymer design, Instron, DMA, ASTM testing, rheology, mechanical characterization, materials design using polymers, HPLC, LC-MS, DLS, UV-Vis Spectroscopy, Mass Spectrometry, ^{1}H NMR, FTIR, TGA, flow cytometry
- Instructional & Project Tools: Adobe Creative Suite, Canvas LMS, Slack, Zoom, Google Workspace
- Software and data analysis: GraphPad Prism, MATLAB, ImageJ, MarvinSketch, TopSpin, Microsoft Office Suite